8x8

EDP 2.0

Redesign
OR
Replatform
OR
Both?



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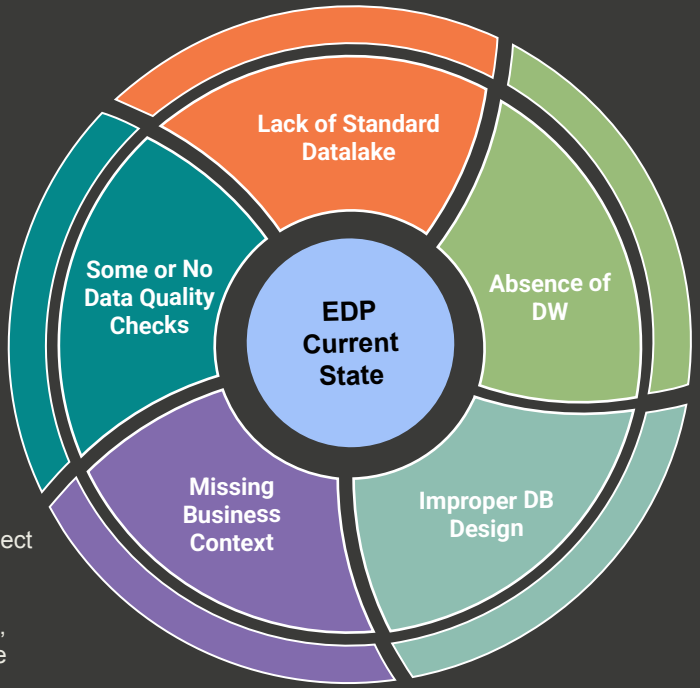
2025

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Key issues in As - Is EDP

- Not having a standardized data lake architecture (e.g., using Amazon S3 as the central raw data repository).
- Splitted ingestion paths — some directly feeding into Snowflake and others through S3, introduces several **downsides** in terms of loss of data lineage and observability, risk of data duplication and inconsistency
- Reduced reusability(S3-based ingestion allows you to **reuse raw data** for different downstream needs (e.g., analytics, ML, audit)

- No data quality checks during or post ingestion
- **Exception Reporting** should be part of data ingestion and not a project on its own.
- No source to snowflake checks for duplicates , data anomalies, cardinality checks.



- No Dimension,Fact,Summary,Semantic Layer tables.Hence the need of projects like **Metrics**
- **Under utilising Snowflake** & **Over Utilizing Tableau** with not pride but with disappointment brought from inconsistency in metric values and poor dashboard performances
- No Data Model
- **SCD Implementation** needs to be revisited and been implemented only for a handful of tables for only SFDC source
- Snowflake default time travel is set to only until 1 day
- Using tableau as a DW engine as compared to a semantic engine
- Snowflake default time travel is set to only until 1 day

- Absence of comprehensive data catalog
- Reverse engineering of EDP produced object level metadata but the business context is missing
- Without business context based dictionary , request for KPI's already existing has to be dealt as new requests

- Creation of Databases /catalog based on projects will keep the chaos adding
- Sticking to Nomenclature based 3-4 layered medallion kind architectures will help support creation of different business objects in a single plane
- performance optimisation best practices missing(like clustering, recommended data types etc)

Lets define some basic tenets..

Redesigning

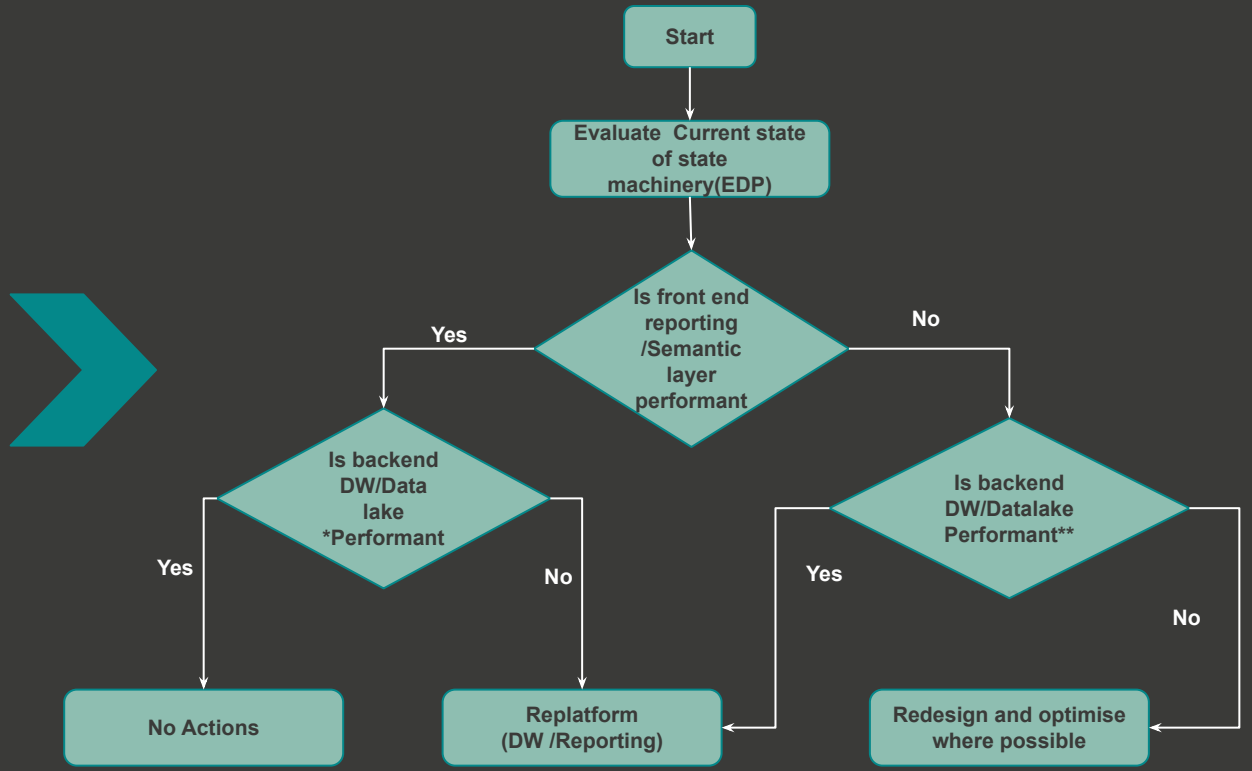
Reedesigning involves improving user experience without changing the underlying platform

- Focus is on Enhancing self serve experience, better decision and business stakeholder engagement
- Goal is to improve user experience & brand perception
- Consider when current design is inflexible & unable to give right data to right people at the right time

Replatforming

Replatforming involves migrating to a new technology platform

- This focuses on migrating to a new technology platform like new ETL /Database/ hyperscaler etc
- Goal is to improve performance and throughput
- When the current platform is outdated, inflexible, or unable to support business growth



★ Is Replatforming alone going to solve our existing problems ?
★ Is Replatforming alone going to add value we are looking for?
★ Can Replatforming enable AI/ML-driven predictive analytics capabilities on the existing enterprise data platform ?
★ Can Replatforming alone enable future vision of empowering our users to “Talk to your Data” using Cortex Analyst (LLM) through Natural language processing (NLP) with the current state of data platform?

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Redesign needs conscious investments

Plan & Design

- Get programme sponsorship
- Define charter and end goals
- Set timelines
- PMO activities(scoping)
- Business stakeholders

Skills & Persona's

- Data Architect
- Data Engineers
- Business Analyst
- IT Leadership / CIO Sponsor
- Project Manager (scoping, timelines)

Tech finalisation

- Agile tools
- Architecture tools
- ELT Routine
- Features in tools
- CI/CD
- Release management

Follow SDLC

- Design
- Develop
- Test
- Deploy
- Monitor & Operationalise

Replatforming would also need most of these but may not solve the business problem

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Risk Compendium..

Risk Category	Risk Name	Description	Impact	Recommendation
Short Term Risks	Poor Query Performance	Raw data is redundant, lacks indexing, & optimized structure. Joins on large tables are slow.	Slower dashboards, frustrated users, delayed insights.	Introduce data modeling and materialized views. Optimize joins.
	Data Complexity	Business users struggle with source-system-specific fields (codes, cryptic names).	Low adoption of data tools, misinterpretation of data.	Implement semantic layer and data dictionary
	No Single version of truth	KPIs are inconsistently calculated across reports	Conflicting business metrics and decision-making	Centralize KPI definitions in a metrics layer or dbt models
	High Transformation Load of Reports	Dashboards carry all logic (joins, calculations).	Reports are slow, error-prone, and hard to maintain.	Push transformations to dbt/Snowflake layers.
	Dependency on Data engineers to explore business data	Business users depend on engineers for understanding or accessing data.	Bottlenecks and slower time to insight.	Empower analysts with well-documented marts and BI training
Long Term Risks	Unscalable Architecture	Schema-on-read and denormalized raw layers degrade at scale.	Performance issues, user frustration, and limited scalability.	Adopt medallion/layered architecture. Refactor for Snowflake best practices.
	Lack of historical tracking	No SCD handling for changing dimensions (e.g., region, product type)	Inability to analyze change over time or see accurate historical views	Introduce SCD Type 2 models using dbt or ETL tools.
	Data Redundancy and inconsistency	No master data model leads to duplication (e.g., customer IDs vary).	Joins fail or yield incorrect results. Trust in data erodes.	Establish a master data strategy and use entity resolution.
	No Governance or Lineage	No visibility into how metrics or tables were built.	Data cannot be audited or trusted.	Implement data catalog, dbt docs, and ETL lineage tools (e.g., Monte Carlo).
	Cost inefficiencies	Repeatedly querying raw tables increases warehouse spend.	Escalating Snowflake bills and redundant compute usage.	Use aggregation layers, caching, and cost-monitoring dashboards.

Conclusion

“Good Things Take Time”, “Quality Comes At A Cost”

What Next?

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Q & A

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